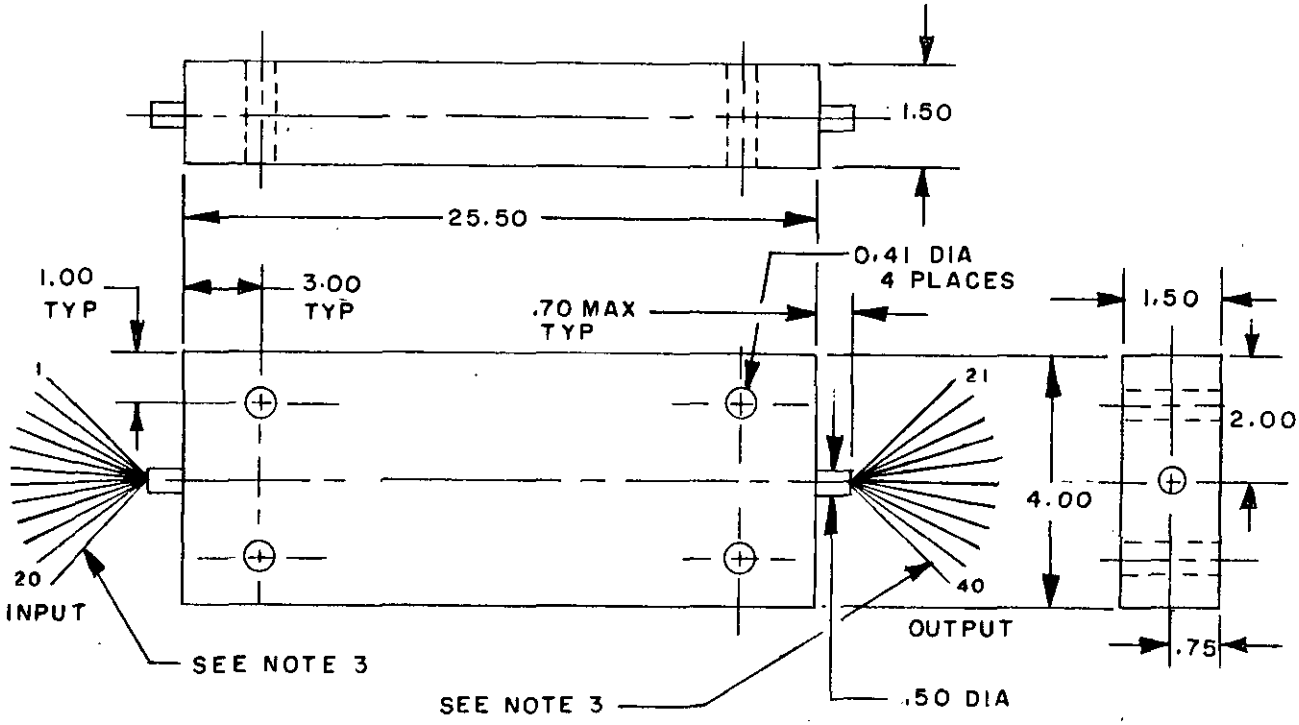
MILITARY SPECIFICATION SHEET

COUPLERS, FIBER OPTIC CABLE, CABLE SPLITTER, PASSIVE, GLASS-PIGTAILED OUTPUT; TYPE II, 820 TO 910 NANOMETERS WAVELENGTH RANGE, 20 X 20 PORT TRANSMISSION STAR, 100/140 AND 50/125 MICROMETER FIBER.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fiber optic couples described herein shall consist of this specification and the latest issue of DOD-C-24621.



cm	Inches	cm	Inches
0.41	.161	1.50	.591
0.50	.197	2.00	.788
0.70	.276	3.00	1.182
.75	.296	4.00	1.576
1.00	.394	25.50	10.047
		95.00	37.430

NOTES:
1. Dimensions are in centimeters.
2. Unless otherwise specified, tolerance is $\pm$ 0.03 centimeter.
3. Pigtails shall be 95 $^{+5}_{-0}$ centimeters long, external to the star coupler package.

FIGURE 1. Fiber optic coupler scale chart.

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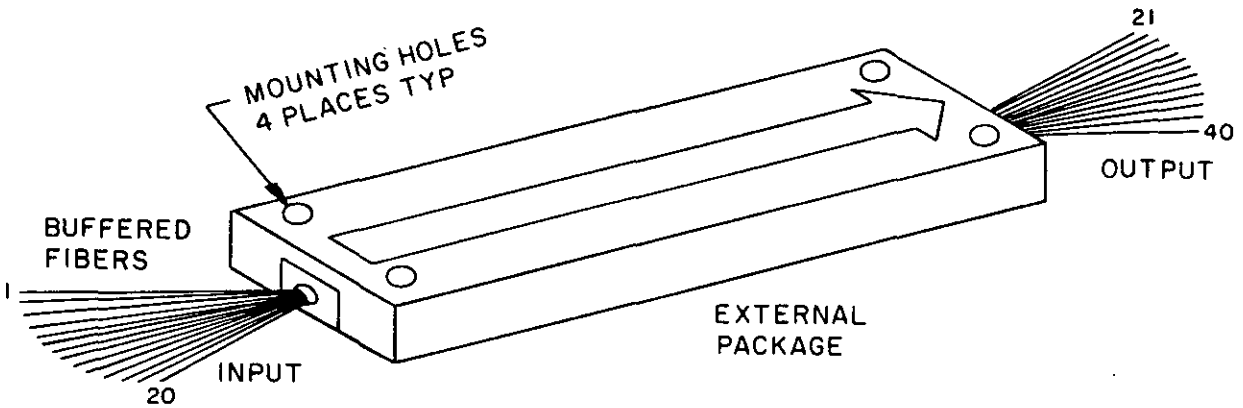
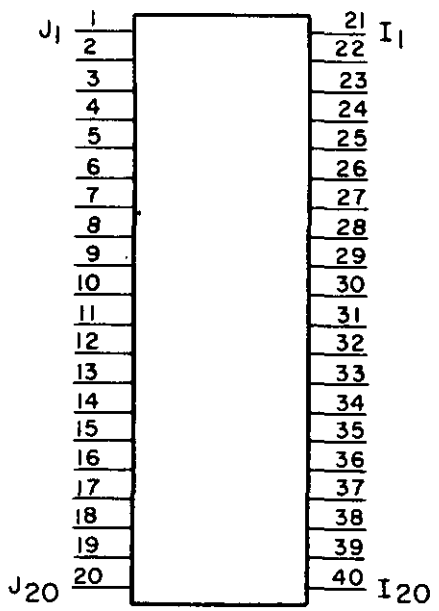


FIGURE 2. Uni-directional coupler.



P(1,21) P(1,22) P(1,23). . . .P(1,40)
P(2,21) P(2,22) P(2,23). . . .P(2,40)
.
.
.
.
P(20,21) P(20,22) P(20,23) . .P(20,40)

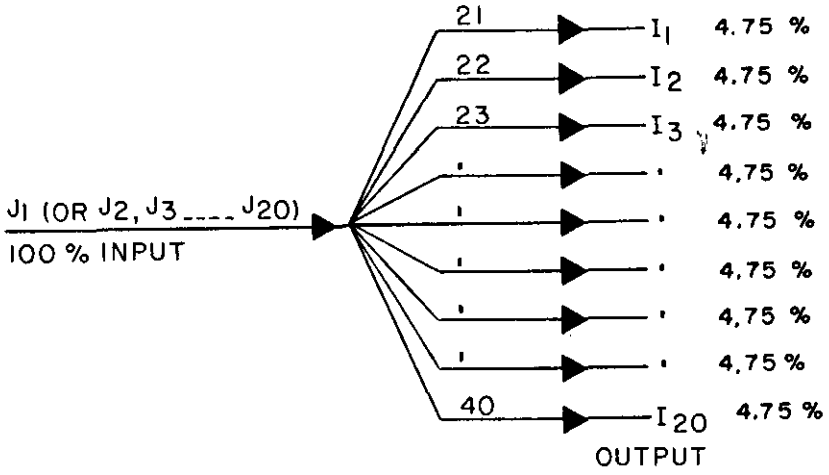
4.75% output 10 $\log_{10}(.0475)$ = -13.2dB

-13.2 -13.2 -13.2 . . . . . .-13.2
-13.2 -13.2 -13.2 . . . . . .-13.2
.
.
.
.
-13.2 -13.2 -13.2 . . . . . .-13.2

NOTES:
1. Ports 1-20 are not useable as outputs when ports 1-20 are being used as inputs and reflected power back to the input ports shall be -40 dB or less.
2. $P_{IJ} = 10 \log_{10} (P_I/P_J)$ where P_J is the input power at port J_1 and P_I is the output power at port I.

FIGURE 3. Transmission matrix derivation configuration.

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NOTES:
1. J_1 and I_1 are illustrated as the dominant ports of the coupler.
2. Each matrix element specified here has a maximum allowed tolerance of ±0.5 dB.
3. Matrix elements above include the connection losses of any splice or mating connector.

FIGURE 3. Transmission matrix derivation configuration - Continued.

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REQUIREMENTS:

Design and construction: See figure 1.

Mass: 1 kilogram maximum.

Pigtail optical fiber cable: All pigtail fibers must be covered and protected by sheath buffer, and coating as for standard cable (see DOD-C-85045). Buffered fiber, kevlar strength member, and outer jacket shall be 2.5 - 3.0 mm diameter per fiber.

Pigtail minimum bend radius: 70 mm maximum.

Splice compatability: In accordance with DOD-C-24623.

Mating cable type: In accordance with DOD-C-85045.

Operating temperature: -40°C to +85°C minimum temperature range.

Dominant input/output port path: (1, 21) 1/.

Designated input port: (J = 1).

Designated output port: (I = 21).

Part number: D24621/3-01 for 100/140 µm fiber; D24621/3-02 for 50/125 µm fiber.

1/ These dominant ports are the ports used to monitor insertion loss for testing purposes (see transmission matrix) (see figure 3).

Note: This coupler shall be unidirectional with the direction of power flow shown by some easily understood means such as an arrow.

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Navy - AS, EC
Air Force - 11, 17, 19, 80, 99
DLA - ES

User activities:
Navy - CG, MC

Preparing activity:
Navy - SH

Agent:
DLA - ES

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